AG3S: Grounding Target Surfaces for Selective Contact in Vision-Language-Action Policies

Anonymous CVPR submission

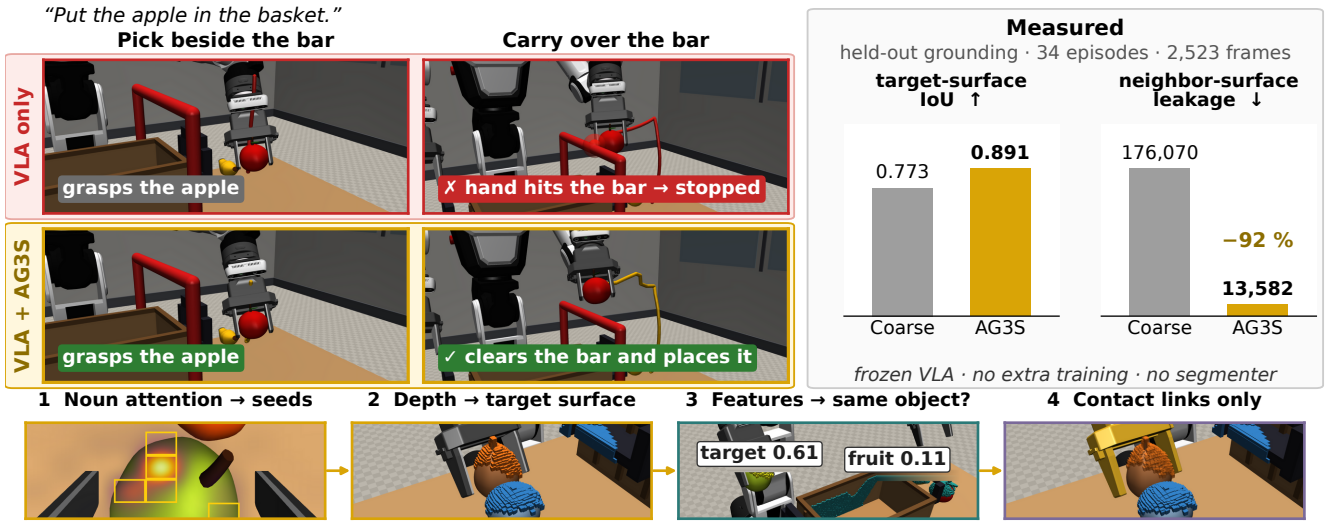


Figure 1. **AG3S-VLA lets a frozen VLA touch the object it names, and nothing beside it.** *Left:* the same policy, scene, seed, and obstacle, recorded on two snapshots of the client. Alone, the policy grasps the apple beside the bar but drives its hand into the bar while carrying and is stopped on contact; with AG3S-VLA it lifts the apple and carries it over. *Bottom*, on a development episode with a pear: noun attention seeds the object, depth continuity recovers its observed surface while the touching fruit stays an obstacle, centered features confirm it is still the same object, and only the designated contact links may enter it (configured rule, illustrated). *Right:* on 34 pre-registered held-out episodes the surface construction improves every episode and, with reassociation, frees 92% less neighboring surface than radius growth; closed-loop results are in Tab. 1.

Abstract

Manipulation requires selective contact: the fingers must touch the instructed object, the wrist must not, and once grasped the object itself must clear the scene. A distance-field safety layer behind a vision-language-action (VLA) policy cannot make these distinctions on its own. Every observed surface is an obstacle, and carving the target out with a ball also erases the fruit touching it. We present AG3S-VLA, which constructs the collision representation a frozen VLA needs from the policy's own evidence. Noun attention decides which *object; depth continuity decides* where *its observed surface ends; centered internal features decide* whether *the selected surface is still the same object; a sub-task probe and physical grasp evidence decide* when *the object turns from target into carried geometry and back. These answers configure link-specific distance queries, so that only designated contact links may enter the accepted target surface while every other link and every neighboring surface stays protected, and a constrained chunk refiner tracks the policy under them. On 34 pre-registered held-out episodes, the surface construction raises observed-surface IoU from 0.77 to 0.88 in every single episode, to 0.89 with temporal reassociation, and removes 92% of the neighboring surface that a radius-grown target would have freed, without an external segmenter or any update to the VLA. In closed loop, the same frozen policy that is stopped by contact with a bar across the carry path in 13 of 14 hurdle episodes and completes none, completes 7 with AG3S-VLA and no bar contact. [TBD: SafeLIBERO and real-robot sentence once results are registered.]*

1. Introduction

Robotic manipulation tasks, such as pick-and-place operations in manufacturing and logistics, require robots

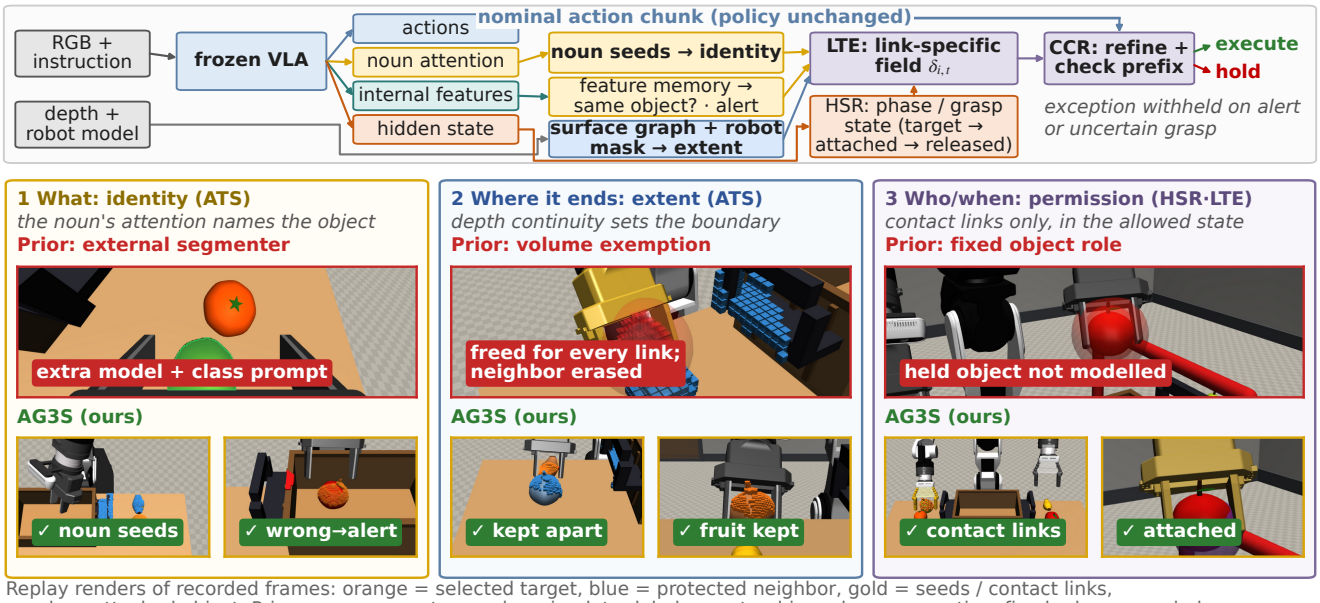


Figure 2. **One frozen VLA decides what may be touched, where that surface ends, and by which link, when.** *Top:* AG3S-VLA reads noun attention, internal features, and hidden state from the same frozen policy that produces the action chunk, and turns them with depth into one link-specific contact representation that constrained refinement (CCR) enforces. *Bottom, each against the closest prior practice. Identity:* eligible noun-attention patches seed the object, and centered features tell whether it is still the same one. *Extent:* depth continuity and a rendered robot mask bound the surface, so the touching fruit stays protected and the fruit between the fingers is kept, where radius growth and a volume mask swallow one and erase the other. *Permission:* only the contact links of the active hand may enter it; after a confirmed grasp the object becomes carried geometry and, on release, an obstacle again. The permission panels are recorded frames of the coarse-extent configuration, which shares this contact interface.

to identify target objects from visual observations and execute appropriate actions according to task instructions. Recent advances in Vision-Language Models (VLMs) have significantly improved visual understanding and language-grounded reasoning by learning semantic relationships between visual observations and natural language. However, semantic understanding alone is insufficient for generating continuous robot actions. To bridge this gap, Vision-Language-Action (VLA) models integrate pretrained vision-language representations with robotic demonstrations to learn policies that directly predict robot actions from observations and task instructions. Representative models, including RT-2 [22], OpenVLA [10], and $\pi_{0.5}$ [13], have demonstrated language-conditioned manipulation capabilities across diverse objects and tasks, as well as promising generalization to unseen environments.

The fundamental capability of VLA models lies in connecting visual observations and language instructions to robot action spaces. Typically, VLA models are trained to predict individual actions or continuous action chunks conditioned on visual observations, robot states, and task instructions. Through this process, these models can implicitly learn not only semantic object information but also spatial relationships and manipulation patterns relevant to task execution. Consequently, VLA models can generate diverse manipulation behaviors without requiring explicitly predefined motion rules or trajectories for each task. **However, the ability to predict actions is fundamentally distinct from the ability to explicitly represent the geometric and physical constraints that those actions must satisfy.** Since conventional VLA policies primarily learn to generate actions consistent with demonstration data, their training objectives neither require an explicit representation of object-level 3D occupancy nor guarantee that the generated actions satisfy robot–environment collision constraints or task-dependent contact conditions.

These limitations become particularly critical in cluttered environments where multiple objects are closely arranged or target objects are partially occluded. For instance, picking an apple adjacent to an obstacle requires the robot not only to identify the intended target but also to distinguish its 3D surface from surrounding objects and determine which regions permit contact and which must be avoided. When multiple visually similar objects are present, 2D regions of interest alone may be insufficient to reliably determine object identity and geometric boundaries. Furthermore, contact with the target object must be permitted during grasping, while the same contact may need to be re-

stricted during other manipulation stages. Therefore, safe manipulation requires both accurate object-level spatial representations and task-dependent contact conditions. **However, task-relevant information encoded in VLA attention maps and hidden states is not directly available as explicit 3D collision constraints.**

Recent studies have explored attention-based target identification [12], obstacle estimation using external perception models [7], and constraint-based action refinement [7, 12] to improve the safety of VLA policies. However, some existing approaches rely on additional segmentation networks to distinguish object regions or approximate complex object geometries using simplified spatial representations [7, 12]. Moreover, even when target and non-target objects can be distinguished, maintaining consistent target identity and geometric boundaries remains challenging in the presence of occlusion and visually similar objects. These challenges motivate a method that directly exploits task-relevant information embedded in VLA representations to segment target object surfaces and transform observed 3D geometry into explicit collision and contact constraints tailored to different manipulation stages.

To address these challenges, we propose **AG3S-VLA**, an Attention-Guided 3D Scene Segmentation framework for generating task-conditioned collision constraints from VLA internal representations. AG3S-VLA identifies task-relevant objects using VLA attention maps and hidden-state features without relying on external segmentation models. Specifically, multi-view RGB-D observations are lifted into a unified 3D spatial representation, where point-wise attention scores and geometric continuity are jointly exploited to segment observed object surfaces. Furthermore, we introduce a spatiotemporal 3D object memory that integrates spatial relationships and temporal feature information to distinguish visually similar objects and preserve target identity under partial occlusion. Based on these representations, AG3S-VLA constructs 3D geometric descriptions of target and non-target objects and defines collision and contact constraints according to manipulation stages and robot links. The resulting constraints are subsequently incorporated into trajectory optimization to refine VLA-generated action chunks while preserving the intended manipulation objective. In this way, AG3S-VLA bridges the gap between implicit task-relevant representations learned by VLA policies and explicit geometric constraints required for safe robotic manipulation.

We evaluate AG3S-VLA on the SafeLIBERO [7] and LIBERO-Safety [3] benchmarks, as well as real-world robotic manipulation tasks. Extensive experiments demonstrate that AG3S-VLA outperforms existing VLA-based methods in manipulation success and safety performance, achieving state-of-the-art results. Real-world evaluations further show strong generalization in cluttered environments involving visually similar and partially occluded objects. These results highlight the effectiveness of leveraging VLA internal representations for 3D spatial perception and explicit collision constraint generation.

Our contributions are threefold:

- **A Novel AG3S-VLA Framework:** We propose AG3S-VLA, a 3D-aware framework that leverages VLA internal representations to generate task-conditioned collision constraints for robotic manipulation. To the best of our knowledge, AG3S-VLA is the first VLA framework to perform task-conditioned 3D object surface segmentation directly from policy-internal representations without relying on external or separately trained segmentation networks.
- **Attention-Guided 3D Segmentation and Object Memory:** We introduce a 3D perception module that segments object surfaces by combining VLA attention scores with geometric continuity. By leveraging hidden-state features and 3D spatial relationships, our method distinguishes visually similar objects and maintains target identity and location under partial occlusion through a spatiotemporal 3D object memory.
- **State-of-the-Art Performance and Real-World Generalization:** Extensive evaluations on SafeLIBERO and LIBERO-Safety demonstrate that AG3S-VLA outperforms existing VLA-based methods in manipulation success and safety performance. Real-world experiments further confirm strong generalization and high success rates in cluttered and occluded environments, validating the effectiveness of VLA-internal 3D perception for task-conditioned constraint generation.

2. Related Work

Vision-Language-Action Models. Recent advances in VLMs have facilitated the development of VLA models that integrate visual perception, language understanding, and robotic action generation. Early approaches, such as RT-2 [22] and OpenVLA [10], formulate robotic control as an autoregressive sequence prediction problem by discretizing continuous actions into tokens. More recent models, including π_0 [2] and $\pi_{0.5}$ [13], employ flow-matching-based action generation and large-scale multimodal training to improve manipulation performance and generalization across diverse tasks and environments. To further enhance spatial understanding, SpatialVLA [14] incorporates 3D positional representations, while PointVLA [11] integrates point cloud features into pretrained VLA policies. Despite these advances, improved spatial representations and action generation do not inherently ensure that predicted actions satisfy explicit geometric collision constraints, particularly in cluttered environments involving close interactions between target objects and surrounding obstacles.

Collision-Aware and Constraint-Guided Manipulation. Several studies have integrated safety mechanisms into VLA policies to mitigate unintended collisions during robotic manipulation. SafeVLA [20] incorporates safety requirements through constrained reinforcement learning, while AEGIS [7] employs vision-language-guided obstacle identification and 3D geometric representations to enforce collision avoidance using control barrier functions. KNOWS [12] instead identifies the manipulation target from the attention of a frozen VLA, while obtaining object geometry from an external segmentation model and approximating each object with an ellipsoid. Other approaches, such as neuro-symbolic safety guidance [4], incorporate collision constraints into flow-matching-based action generation, enabling predictive trajectory correction. VLA-Stem [23] further introduces a lightweight safety policy that adapts VLA-generated motion references to spatial constraints and dynamic obstacles. These methods demonstrate the effectiveness of safety-aware action generation and correction, highlighting the importance of reliable geometric representations for enforcing collision constraints. Frozen VLA representations have also been shown to encode task progress and failure signals [1, 6], suggesting that policy-internal features carry task-relevant information beyond action prediction.

In parallel, several studies have investigated language-grounded geometric representations for robotic manipulation. VoxPoser [8] constructs 3D value maps from language instructions and visual observations to guide model-based motion planning. ReKep [9] represents manipulation tasks through relational constraints over 3D keypoints, enabling optimization-based action generation. Other works, such as LMAffordance3D [21], ground language-specified interaction regions on 3D object point clouds. Although these approaches provide spatial guidance for robot planning, constructing collision constraints for manipulation additionally requires distinguishing the intended target from nearby objects and determining which contacts are permissible throughout task execution. Motivated by this distinction and unlike KNOWS, which uses attention only to select among externally segmented objects, our work leverages internal VLA attention and hidden-state features together with multi-view RGB-D observations to identify task-relevant 3D surfaces and construct geometric collision constraints with task-dependent contact permissions. These constraints are designed to support downstream trajectory optimization while preserving the manipulation objective.

3. Method

In this section, we present AG3S-VLA, which turns the task-relevant information inside a frozen VLA into explicit 3D collision and contact constraints. After formulating the problem (Sec. 3.1), we describe the two core components, attention-guided 3D segmentation (Sec. 3.2) and a spatiotemporal 3D object memory (Sec. 3.3), followed by the stage- and link-specific constraints they define (Sec. 3.4) and the refinement of the action chunk under them (Sec. 3.5).

3.1. Problem formulation

Given a language instruction l, RGB observations I_t, and the robot state q_t at request t, a VLA policy π_θ predicts an action chunk

$$A_t = [a_t, \dots, a_{t+H-1}] = \pi_\theta(I_t, q_t, l), \tag{1}$$

where H is the chunk length. Since π_θ is trained to reproduce demonstrations, A_t carries no guarantee that the robot clears the scene, and in clutter the constraint it must satisfy is itself task-dependent. For a collision sphere i of the robot or of a grasped object, with centre $c_i(q)$, radius r_i, and margin μ_i, the clearance at waypoint k reads

$$h_{ik}(q_k) = \delta_{i,t}\big(c_i(q_k)\big) - r_i - \mu_i \geq 0, \tag{2}$$

where $\delta_{i,t}$ is the signed distance field that sphere i queries. A single field computed from all observed surfaces forbids the grasp itself, because the fingers must touch the target. The field therefore depends on three quantities that π_θ never outputs explicitly: the observed surface $\mathcal{T}_t$ of the target object, the manipulation stage s_t, and whether sphere i belongs to a contact link in $\mathcal{L}$, the palm and fingers of the manipulating hand. Prior safety layers take the target region from an external segmenter or a primitive such as an ellipsoid [7, 12], and the latter also frees the neighbor that touches the target.

AG3S-VLA instead derives $\mathcal{T}_t$ from the policy's own forward pass, using in addition only calibrated depth and a robot model covered by collision spheres. Fig. 3 shows the components and Alg. 1 the order in which they run at each request: the attention of the target noun is grown along depth into a candidate surface (*target segmentation*, Sec. 3.2), a 3D object memory decides whether it is the target (Sec. 3.3), the stage selects which surfaces each link may enter (Sec. 3.4), and the chunk is refined under Eq. (2) (Sec. 3.5).

3.2. Attention-guided 3D segmentation

As discussed in Sec. 1, 2D regions of interest cannot separate touching or visually similar objects. VLA attention indicates *which* object the instruction refers to, but only at the resolution of image patches, whereas depth reveals *where* surfaces end but not which of them is the target. To this end, we lift the observations into a unified 3D representation and jointly exploit point-wise attention scores and geometric continuity (target segmentation in Fig. 3).

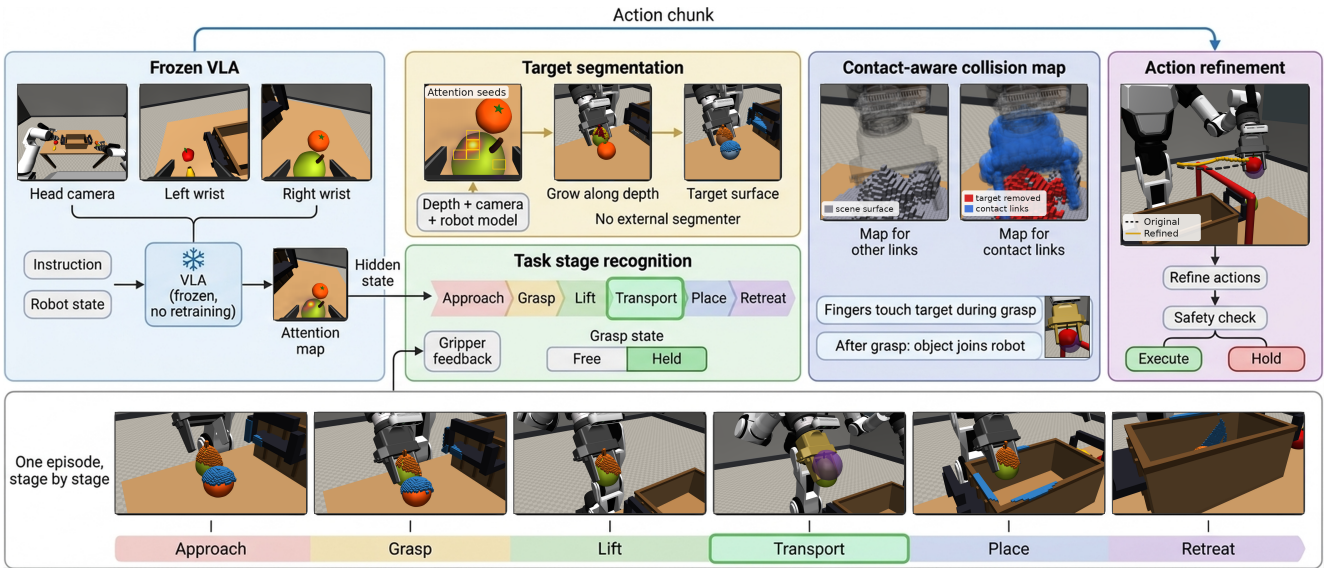


Figure 3. **Method overview.** A frozen VLA predicts the action chunk from three camera views, the instruction, and the robot state; the same forward pass exposes the noun attention map and the hidden state. **Target segmentation** (Sec. 3.2) grows the attention seeds along depth into the target's observed surface (Eq. (5)), with no external segmenter. **Task stage recognition** (Sec. 3.4) classifies the stage of the task from the hidden state, and the grasp state follows from gripper feedback. The **contact-aware collision map** (Sec. 3.4) gives the contact links a map from which the target surface is removed and every other link the full scene; once the object is grasped it joins the robot. **Action refinement** (Sec. 3.5) refines the chunk under Eq. (2) and executes it or holds. The bottom strip shows one simulated episode stage by stage.

Algorithm 1 AG3S-VLA at request t

Require: I_t, depth, q_t, l, memory $\mathcal{M}$, grasp state z, last target $\bar{\mathcal{T}}$

$A_t, a \leftarrow \pi_\theta(I_t, q_t, l)$ ▷ chunk, noun attention
$\mathcal{G} \leftarrow$ surface graph of robot-masked depth
$m^* \leftarrow (\Lambda + L_{\mathcal{G}})^{-1}\Lambda y$, seeds of Eq. (3)
$R_t \leftarrow$ region of m^* with most seed mass
update candidates and attention lead in $\mathcal{M}$
if $z \in \{\text{CLOSING}, \text{HELD}\}$ **then**
  $\mathcal{T}_t \leftarrow$ pre-closure surface, moved with it
else if R_t overlaps the target in $\mathcal{M}$ **then**
  $\mathcal{T}_t \leftarrow R_t$
else
  $\mathcal{T}_t \leftarrow \bar{\mathcal{T}}$ ▷ abstain
end if
$z \leftarrow$ grasp state from finger opening
$s_t \leftarrow$ stage from z and palm–target distance
$d_t,\ d_t^{\setminus\mathcal{X}} \leftarrow$ ESDFs of the TSDF, Eq. (6)
attach $\mathcal{T}_t$ as spheres if $z = \text{HELD}$
$Q^* \leftarrow$ SQP on Eq. (8) with $\delta_{i,t}$ of Eq. (7)
execute Q^* if all $h_{ik} \geq 0$, else HOLD

Unified 3D representation. The depth pixels of all cameras are lifted into the robot frame and voxelized into surface nodes with centroids x_i and normals n_i; pixels on the robot's collision spheres are discarded, and a RANSAC support plane is marked as support. Each node keeps the image patches its pixels come from, so attention transfers to 3D without re-projection.

Point-wise attention scores. Let a_p be the attention from the instruction's target-noun token to image patch p in one fixed head of the frozen VLA, selected once on development episodes. Copying each patch score to every point under it would let large, distant surfaces outweigh the small object the noun refers to. We instead keep the top-ranked patches $\mathcal{P}$ and distribute the normalized score of each patch over the nodes it observes in proportion to pixel share,

$$\lambda_i^+ = \sum_{p \in \mathcal{P}} \frac{N_{ip}}{N_p} \frac{a_p}{\max_{r \in \mathcal{P}} a_r}, \tag{3}$$

where N_p counts the retained pixels of patch p and N_{ip} those on node i. Each patch thus contributes one unit of evidence regardless of the number of points it covers.

Geometric continuity. A radius graph would join two touching objects. We connect two nodes only when neighboring pixels of the same image link them without a depth discontinuity, so that proximity alone cannot bridge a visi-

ble depth break, and weight each admitted edge by

$$w_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{\sigma_x^2} - \frac{1 - |n_i^\top n_j|}{\sigma_n}\right). \quad (4)$$

Attention-guided segmentation. Let $\lambda_i^- = 1$ on support nodes, $\Lambda = \mathrm{diag}(\lambda^+ + \lambda^-)$, and $y_i = \lambda_i^+ / \Lambda_{ii}$. With the Laplacian $L_{\mathcal{G}}$ of the surface graph $\mathcal{G}$, the target membership $m^* \in [0,1]^{|\mathcal{G}|}$ is

$$m^* = \arg\min_m (m-y)^\top \Lambda (m-y) + m^\top L_{\mathcal{G}}\, m, \quad (5)$$

obtained from $(\Lambda + L_{\mathcal{G}})m^* = \Lambda y$ on the GPU. Attention decides which surface receives evidence, and geometric continuity decides how far it flows; components without a seed stay unknown, never free. Thresholding m^* and keeping the connected region with the greatest seed mass yields the candidate surface R_t. The target surface thus comes from the policy's own attention and the scene geometry alone, without an external or separately trained segmentation network.

3.3. Spatiotemporal 3D object memory

The segmentation fixes the extent of a surface at one request, but not which object is the target across requests: attention can move to a similar object, and the approaching hand occludes and pushes the target exactly when contact is imminent. We therefore keep the target in a memory of 3D objects that integrates their spatial relationships with their history over time.

Spatial relationships. The lifted attention is grouped into 3D-connected candidates above the support. Candidates floating above it or forming the destination, such as an obstacle bar or the basket, are filtered out, and one that merges two touching objects is split by its attention share.

Temporal association. A candidate becomes, or replaces, the target only after leading the attention for several consecutive requests, and R_t is accepted only if it overlaps that target (Alg. 1, lines 6–12). An occluded target keeps its last observed geometry, and a grasped one keeps its pre-closure geometry while the hand pushes it. The memory thereby separates *identity*, decided over time, from *extent*, decided at each request, and missing evidence can only keep or withdraw an accepted surface, never extend it.

3.4. Task-conditioned collision constraints

Contact with the target must be permitted while grasping, restricted in the other stages (Sec. 1), and granted only to the links meant to touch it. We now build $\delta_{i,t}$ of Eq. (2) accordingly.

Task stage recognition. Since appearance cannot tell a gripper holding the object from one closed on nothing, a grasp state z_t becomes HELD only when the measured finger opening stalls above the commanded width, and PLACED when the gripper reopens or the object slips out. With the palm–target distance, z_t yields the stages of Fig. 3 from approach to retreat without any learned component beyond the frozen VLA; the thresholds are given in the supplementary material.

Target-removed map. The scene map d_t is a Euclidean signed distance field (ESDF) computed with cuRobo [16] from a TSDF of the robot-masked depth, coarse over the workspace and fine around the hand. For the contact links, we compute a target-removed map $d_t^{\setminus\mathcal{X}}$ from the same TSDF without erasing any voxel, by dropping the surface voxels $\mathcal{X}_t$ owned by the target from the seeds of the distance transform,

$$\mathcal{X}_t = \{x :\ d_T(x) < d_N(x)\} \cup \{x :\ d_T(x) \le \epsilon_c\} \cup \mathcal{H}_t^\omega, \quad (6)$$

where d_T and d_N are the distances to the nearest sample of $\mathcal{T}_t$ and of any other observed surface. The first set keeps every seed closer to another surface, so the touching neighbor still seeds the map. A contact allowance ϵ_c around the target, together with the convex hull $\mathcal{H}_t^\omega$ of $\mathcal{T}_t$ applied just before the grasp, covers the unobserved back and interior of the target. Seeds below the target's bottom are kept to protect the support.

Link- and stage-specific queries. Each sphere queries

$$\delta_{i,t} = \begin{cases} d_t^{\setminus\mathcal{X}}, & i \in \mathcal{L} \ \wedge\ s_t \in \{\mathit{approach}, \mathit{grasp}\}, \\ d_t, & \text{otherwise}, \end{cases} \quad (7)$$

so only the fingers and palm of the manipulating hand may reach the target. The clearance of Eq. (2) is imposed on the spheres of both palms and fingers; the forearms, the torso, and self-collision are not constrained. Once z_t = HELD, the target joins the robot as attached spheres that query the environment, and its surface stops seeding the scene map; after release it is an obstacle again. Therefore, one observed scene yields different constraints for different links and stages, which a single collision map cannot express.

3.5. Constrained action refinement

Finally, trajectory optimization refines the chunk under these constraints while preserving its manipulation intent. With the reference waypoints Q^{ref} of A_t and the deflection $D = Q - Q^{\mathrm{ref}}$, we solve

$$\min_{Q,\,\xi \ge 0} \ \|W_t D\|_F^2 + w_{\mathrm{s}}\|\Delta^2 Q\|_F^2 + w_\xi \textstyle\sum_{i,k} \xi_{ik} + \mathbb{1}_t^{\mathrm{g}}\, w_{\mathrm{d}}\|\Delta D\|_F^2$$
$$\text{s.t. } h_{ik}(q_k) \ge -\xi_{ik} \ \ \forall i,k, \qquad Q \in \mathcal{Q}, \quad (8)$$

by sequential quadratic programming with OSQP [15], where W_t is a diagonal joint weighting, lower for the wrist joints near the target, $\mathcal{Q}$ holds joint and motion limits, and w_ξ dominates. The deflection-rate term acts only after the grasp ($\mathbb{1}_t^{\mathrm{g}}$), keeping an avoidance deflection from snapping back at chunk boundaries without pulling the hand off the grasp pose before it. A continuity term to the previously executed chunk is gated in the same way. A failed safety check on the nonlinear clearance makes the robot HOLD (Alg. 1, line 18), which we count as an intervention, not a success.

4. Experiments

We ask four questions. **Q1** Does AG3S-VLA let a frozen VLA complete tasks it otherwise fails through contact, and at what cost elsewhere (Sec. 4.2)? **Q2** How accurately does the policy's own attention recover the target's observed surface without a segmenter (Sec. 4.3)? **Q3** Which internal signal should decide which part of the problem (Sec. 4.4)? **Q4** What does it cost at run time (Sec. 4.5)? SafeLIBERO (Sec. 4.6) and a real robot (Sec. 4.7) follow.

4.1. Setup

Platform and policy. Our main testbed is a dual-arm RB-Y1 in MuJoCo [17] performing instructed fruit pick-and-place ("Put the pear in the basket") among distractor fruit, with a frozen, embodiment-adapted $\pi_{0.5}$ [13] (two 7-DoF arms, two gripper channels) that is never updated. AG3S-VLA receives three calibrated RGB-D views and the robot model. Scenes come in four obstacle profiles: *clear*; *adjacent*, a post 10 cm from the target in the pre-registered set and closer in the failure study of Fig. 5; *place*, a post beside the basket rim; and *hurdle*, a bar across the carry path.

Protocol and definitions. Closed-loop arms run the same episode with the same policy seed; episode lists and seeds are fixed from metadata before any rollout (hash recorded), and every failure stays in the denominator. Contacts are measured by replaying recorded states in the simulator, never by the controller's own verdicts, and classed as robot–obstacle, carried-object–obstacle, hand–neighbor, non-contact-link–target, and the allowed hand–target. An episode ends at its step limit or at the first robot–obstacle contact (an *obstacle stop*). *Task success*: object in the basket and released. *Safe success*: additionally no forbidden contact before release (hurdle set: no robot or object contact with the bar; pre-registered set: also no non-contact-link–target or neighbor contact). A *hold* is one chunk in which the prefix check fails and the arm waits. Grounding uses a 5 mm voxel grid with soft IoU, frame metrics averaged per episode and then across episodes, and 2,000 episode-paired bootstraps. *Neighbor leak* is the pooled false-positive voxel mass on neighboring objects relative to radius growth (a ratio of sums). The held-out grounding set is the 34 of 43 collected episodes whose collection rollout succeeded; the 9 failures are analysed in the supplement. All parameters were fixed on development data and frozen beforehand (supplement, Tab. S1): noun attention from prefix layer 10, head 7; identity descriptors from prefix layer 8; the probe on layer-4 key/value features; $H = 50$, $J = 16$, $K = 8$.

Table 1. **Closed-loop results.** Safe = safe success; Hold = hold chunks summed over runs. Hurdle: 7 episodes with 2 seeds each. VLA-only hurdle runs were recorded on an earlier client snapshot (same episodes, seeds, and obstacles), so no discordant counts are given for that pair; ours vs. Coarse: both 6, Coarse only 5, ours only 1. Pre-registered: [TBD: 40] scenes (clear 14, adjacent 13, place 13) chosen from metadata only.

Scenes	Method	Success	Safe	Hold
Hurdle (14)	VLA only	0/14	0/14	–
	Coarse	11/14	11/14	82
	Ours	8/14	7/14	11
Pre-reg. (40)	VLA only	[TBD:]	[TBD:]	–
	Coarse	[TBD:]	[TBD:]	[TBD:]
	Ours	[TBD:]	[TBD:]	[TBD:]

Compared configurations. *VLA only* runs the frozen policy without any geometric layer. *Coarse* is the previous AG3S-VLA grounding with the same noun attention: 1 cm fusion, 3 cm radius growth with DBSCAN [5], the sphere/capsule robot filter, and a ball-shaped exemption with ownership splitting, sharing fields, refiner, and checks with ours. *Surface (vol. mask)* is our graph with the sphere/capsule filter; *Surface* adds the rendered mask; *Surface + ID (ours)* adds reassociation and the alert. Ours thus differs from Coarse in extent and robot mask at once; no closed-loop arm with our surface and the volume mask has been run. [TBD: External segmenter (Grounded-SAM / YOLOE [18]) masks lifted with the same depth) on the held-out set.]

4.2. Q1: closed-loop execution

Where the policy cannot see geometry it collides; with AG3S-VLA it passes. Under a protocol that ends an episode at the first robot–obstacle contact, the frozen policy completes none of the 14 hurdle episodes: it grasps the apple in 11, but its hand meets the bar during the carry and the episode is stopped in 13 (Tab. 1). Nothing in the policy represents the bar, so a better target estimate would not help. With AG3S-VLA the same policy completes 7 episodes without touching the bar before release; the Coarse configuration completes 11. The carry rests on attached geometry: in the coarse-extent lineage of the same contact interface, carried spheres removed every carry-phase apple–bar contact (5 of 14 runs to 0; supplement).

Table 2. **Held-out surface grounding.** 34 episodes, 2,523 frames, all settings frozen beforehand. Gain = paired IoU gain over Coarse with 95% CI; Surface improves 34 of 34 episodes, Surface + ID 32 of 34. Leak = neighbor leak relative to Coarse.

Extent	IoU	Prec.	Leak	Gain [95% CI]
Coarse (radius growth)	.773	.894	1.00	–
Surface (vol. mask)	.832	.955	0.26	–
Surface	.876	.949	0.26	+.103 [.083, .123]
Surface + ID (ours)	**.891**	**.966**	**0.08**	+.118 [.083, .151]
External segmenter	[TBD:]	[TBD:]	[TBD:]	[TBD:]

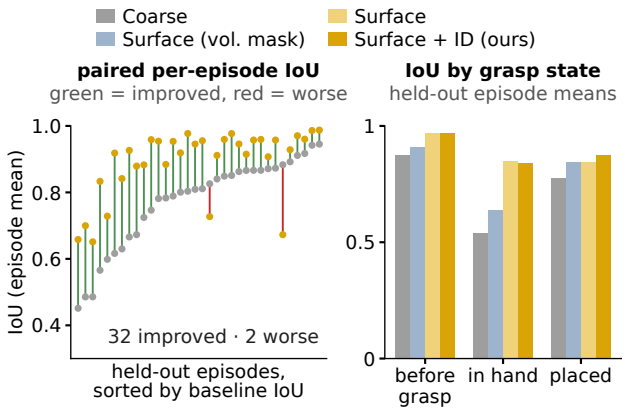


Figure 4. **Held-out grounding.** *Left:* per-episode IoU of Coarse (grey) and ours (gold), one connector per episode; 32 of 34 improve, 2 worsen. *Right:* IoU by grasp state. The rendered robot mask matters most while the fruit is in the hand, where a volume mask erases it.

The richer surface is not yet a higher success rate. Against Coarse under the same noun attention, ours reaches safe success in 7 rather than 11 episodes (both 6, Coarse only 5, ours only 1; exact $p = 0.22$); five of its six failures are missed grasps, one a slip. It holds in 11 chunks against 82, touches the bar in 3 runs against 2, and once lets a non-contact link touch the target; fewer holds here read as less conservative, not as better. Since the arms differ in extent and robot mask at once, the records do not isolate a cause: on 34 held-out seeds chosen because Coarse had succeeded, ours retained 28, and reruns of the six failures with the simulator's visible target surface rescue 2, with the sphere mask 2 others (supplement). The coarse configuration shows the symptom the surface extent targets, a hand–neighbor contact before release in 4 of 16 serving-test episodes; whether the precise extent removes it without losing grasps is what the pre-registered set measures. [TBD: Fill from IFG-CL3 and the VLA-only run on the same 40 scenes; report whichever direction the result takes.] In clear scenes an earlier integrated snapshot neither helped nor hurt the policy (30 vs. 29 of 48; 8/7 discordant).

4.3. Q2: grounding the observed target surface

The surface construction improves every held-out episode. The two registered paired gains, the surface graph (+0.103 [+0.083, +0.123] over Coarse) and the rendered robot mask on top of the same graph (+0.044 [+0.038, +0.050]), both hold in 34 of 34 episodes (Tab. 2); with reassociation the gain is +0.118 [+0.083, +0.151], improving 32 episodes and worsening 2 (Fig. 4). The gain is largest where contact decisions are made: while the fruit is in the hand, IoU rises from 0.640 under the volume mask to 0.848, because the mask no longer erases the object between the fingers; before the grasp the surface is nearly exact (0.970).

What is removed matters more than what is kept. Precision, not recall, carries the improvement: radius growth reaches the target but grows into whatever touches it. Surface continuity removes 74% of the neighboring surface that radius growth would have freed, and reassociation 92%, because a drifted selection is replaced by the remembered object instead of being carved out. Reassociation changes IoU little (+0.015 [−0.012, +0.039], 14 better, 6 worse; the two worsened episodes are wrong first references, Sec. 5); its value is in what it refuses to free. The development set shows the same ordering (0.837 to 0.893, 27 of 32 episodes).

4.4. Q3: which signal decides what

Internal features do not refine a boundary. With robot removal and seeding fixed on the development set, adding feature affinities to the graph leaves IoU unchanged across projector and prefix layers 4, 8, and 12 (0.893 to 0.893–0.894); using features as unaries lowers it to 0.79–0.83 for every representation tried, including SigLIP [19]; feature-conditioned region growing (0.755) trails geometric growing (0.832); and shuffling the features drops the combined method to 0.485 (supplement, Tab. S3). The reason is spatial: 42% of target nodes fall in patches that are mostly not target, and 74% of the edges crossing a measured object boundary lie inside a single patch, so a 16×16 descriptor has no value change where the surface changes. The descriptors are informative, but about a different question.

They do tell whether it is still the same object. Raw patch features score every region in a scene alike (cosine $\approx$ 0.97–0.98 to the remembered target); centered against the frame mean they separate the target from the nearest alternative (0.61 vs. 0.42 in the frame of Fig. 3). As a consistency signal on held-out frames, chain agreement flags 85 of 100 wrong selections at the frozen threshold (AUROC 0.925; 0.924 when the failed rollouts are included); emptying flagged regions removes 88% of wrongly selected surface mass at a 6.7% loss of true target mass. This is the

Table 3. **Per-request latency (ms, median / p95)** on the 34-seed held-out closed-loop runs of Sec. 4.2, 2,516 requests per arm; on the hurdle runs ours measures 3,331 / 4,979. The policy alone answers in 199 ms. Chunks of 50 steps with 8 executed give a 533 ms nominal period.

Stage	Coarse	Ours
Client round trip	1,137 / 1,924	2,858 / 3,623
Server total	796	2,587
surface graph, region, labels	–	1,663
rendered robot mask	–	159

Table 4. **SafeLIBERO.** Task success (TSR) and collision-avoidance rate (CAR) following the benchmark's definitions [7]; Level I places the obstacle beside the target, Level II across the path. Upper block: same checkpoint, our runs. Lower block: numbers as reported by each paper, with their own checkpoints and protected-link scope. [TBD: Fill from the SafeLIBERO run.]

	Level I		Level II	
Method	TSR	CAR	TSR	CAR
Frozen policy only	[TBD:]	[TBD:]	[TBD:]	[TBD:]
AG3S-VLA (ours)	[TBD:]	[TBD:]	[TBD:]	[TBD:]
VLSA (reported) [7]	[TBD:]	[TBD:]	[TBD:]	[TBD:]
KNOWS (reported) [12]	[TBD:]	[TBD:]	[TBD:]	[TBD:]

division of labour the method is built on: geometry for extent, features for identity.

Geometry also sharpens the task state. Adding target–hand distance, height, and region size to the sub-task probe and gripper state raises leave-one-episode-out phase accuracy on simulator phase labels from 0.878 to 0.939 (paired episode gain $+0.067$ $[0.015, 0.118]$), and held-state accuracy from 0.828 for a gripper threshold to 0.933; the probe alone (0.783) mostly confuses *home* with *place*.

4.5. Q4: run time

Both configurations exceed the nominal chunk period in our unoptimized prototype, and ours costs 2.5× more, almost entirely in the per-request surface graph (Tab. 3); the graph is built on the CPU and the backbone copy is not shared, both engineering steps we have not taken.

4.6. SafeLIBERO

SafeLIBERO adds obstacles to the LIBERO suites and scores a frozen $\pi_{0.5}$ with and without a safety layer [7]. We run the same checkpoint with AG3S-VLA on [TBD: suites, levels, episodes per task]; the policy outputs end-effector displacements, which CCR refines through the robot's Jacobian, and the sub-task probe is replaced by the gripper rule because LIBERO demonstrations carry no phase labels. [TBD: Result sentences: TSR/CAR per level, protected-link scope, and how the reported methods differ in checkpoint and metric definition.]

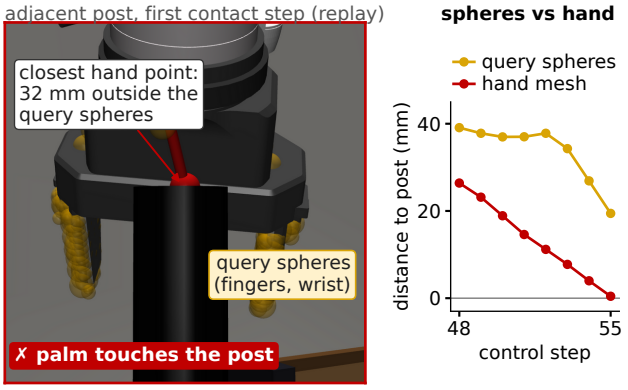


Figure 5. **Failure case: coverage, not grounding.** With a post directly beside the target, the palm rests on the post at the first contact (red point) while every constraint sphere stays clear of it; the hand point closest to the post lies 32 mm outside the sphere union. The frozen policy alone makes the same contact.

4.7. Real robot

[TBD: Hardware (RB-Y1), cameras and calibration, tasks and object sets, number of trials per condition, success and contact outcomes, latency on the robot; pending.]

5. Limitations and Conclusion

Our closed-loop evidence is in simulation on one robot and four objects, with replayed depth for the offline study; SafeLIBERO and the real robot (Secs. 4.6–4.7) widen this but do not remove it. The precise surface does not yet convert into higher success than the coarse exemption on the hurdle set, and the cause is not isolated: our arm changes extent and robot mask together, and neither the simulator's own target surface nor the volume mask alone recovers the failures. Whether the identity alert changes any closed-loop outcome is unmeasured; in the held-out runs it withheld 43 chunks whose outcome we have not attributed. The prototype runs at about 0.35 Hz chunk rate (2.9 s per request); the dominant cost, the per-request surface graph, is amenable to caching and GPU construction, and the duplicated backbone pass can be shared. We reconstruct observed surfaces only: the scene field treats unobserved space as free, so occluded geometry, missing depth, and sub-voxel contacts remain unresolved, and robot-depth matching needs accurate meshes and kinematics. Query coverage is bounded by the sphere model: with a post directly beside the target, both the frozen policy and AG3S-VLA touched it because the real hand surface lay 13–39 mm outside the modeled spheres (Fig. 5). The reassociation can preserve a wrong first reference, which is what the two worsened held-out episodes show, and the supervised phase model uses effectively floor-referenced height. A hold is an intervention, not a safety certificate.

A frozen VLA already says which object it is about to touch and what stage of the task it is in; it does not know where that object's surface ends or which of its own links may enter it. AG3S-VLA keeps these questions apart and answers each with the evidence that can: attention seeds, depth continuity bounds, centered features re-identify, physical feedback authorizes, and link-specific fields carry the answers into constrained execution, without an external segmenter or any change to the policy. We see this separation of evidence as what lets geometric safety layers and learned policies share a scene rather than one overruling the other.

LLM Usage Statement

We used Anthropic Claude models through the Claude Code assistant in three parts of this work. *Manuscript:* drafting and editing the text of all sections, including wording and grammar, under the authors' direction; the authors checked every claim, number, and citation against the experiment records and the cited papers. *Figures:* writing the plotting and composition code that renders the figures from recorded simulation frames and result files; no image content was generated by a model. *Software and experiments:* writing and testing parts of the research code and scripting experiment runs, result aggregation, and evidence audits; the method design, experimental decisions, and interpretation of results are the authors'. [TBD: Authors: confirm this list and add any other generative tool used, e.g. for figure-layout drafts.]

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